



StudentHub

GRADUATION / SOFTWARE ENGINEERING PROJECT

The all-in-one **campus platform** for university students

One product connecting students, activity agents, and administrators — web, mobile, and a unified API.

Team: Abdelrahman Yahia (Lead), Yousef Alaa, Mohamed Ezz, Marwan Mohamed, Omar Mohamed, Ali Haggag, Mahmoud Abdelhady, Omar Abdeen

Stack: React · .NET (Clean Architecture) · Flutter · SQL Server Date: June 2026

Campus life is **scattered everywhere**

Students juggle a different tool for every need:

- Activities & events live on **social media posts** and group chats
- Applications happen over **scattered forms** with no tracking
- Achievements and engagement are **invisible** — no portfolio, no proof
- Services, housing, and community are **fragmented** across channels



No single source of truth

Students miss opportunities; agents can't reach the right audience; admins have no oversight.



No recognition loop

Participation isn't measured or rewarded, so engagement stays low.

OUR SOLUTION

One platform. Three roles. Zero friction.



Students

Discover & apply to activities, earn points & badges, build a public portfolio, shop the store, and join communities.



Agents

Publish activities, manage applicants on a kanban board, verify check-ins, and track performance with analytics.

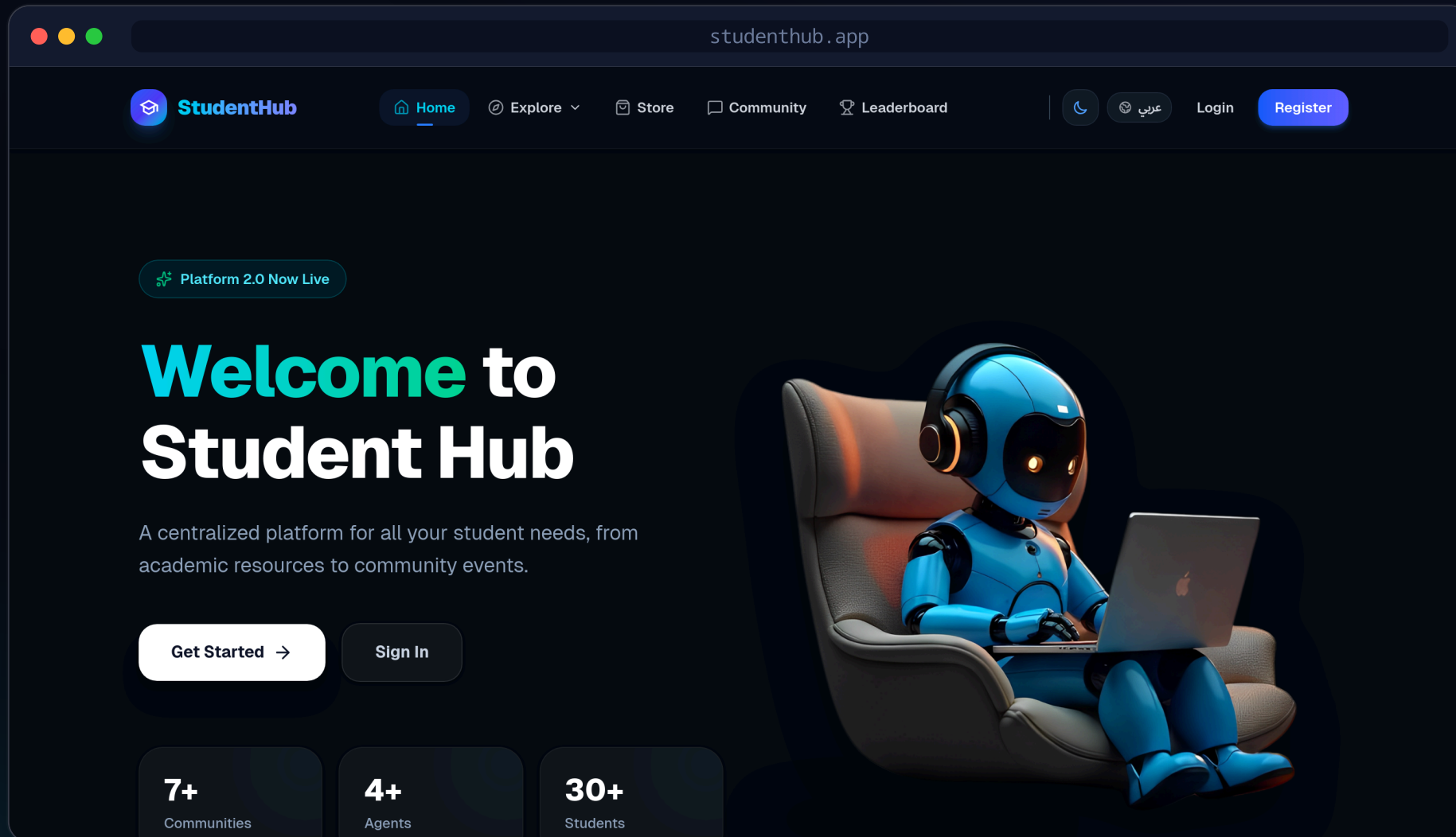


Admins

Approve agents, moderate content, oversee finance, manage users, and maintain platform-wide lookups.

Backed by **role-based access control** and a single **REST API** shared by web & mobile.

This is real, running software



WHAT WE BUILT

Four pillars, **one experience**



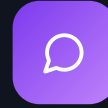
Activities

Browse, filter, apply, and track decisions. Agents manage applicants end-to-end.



Engagement

Points, badges, leaderboards, and a shareable student portfolio.



Community

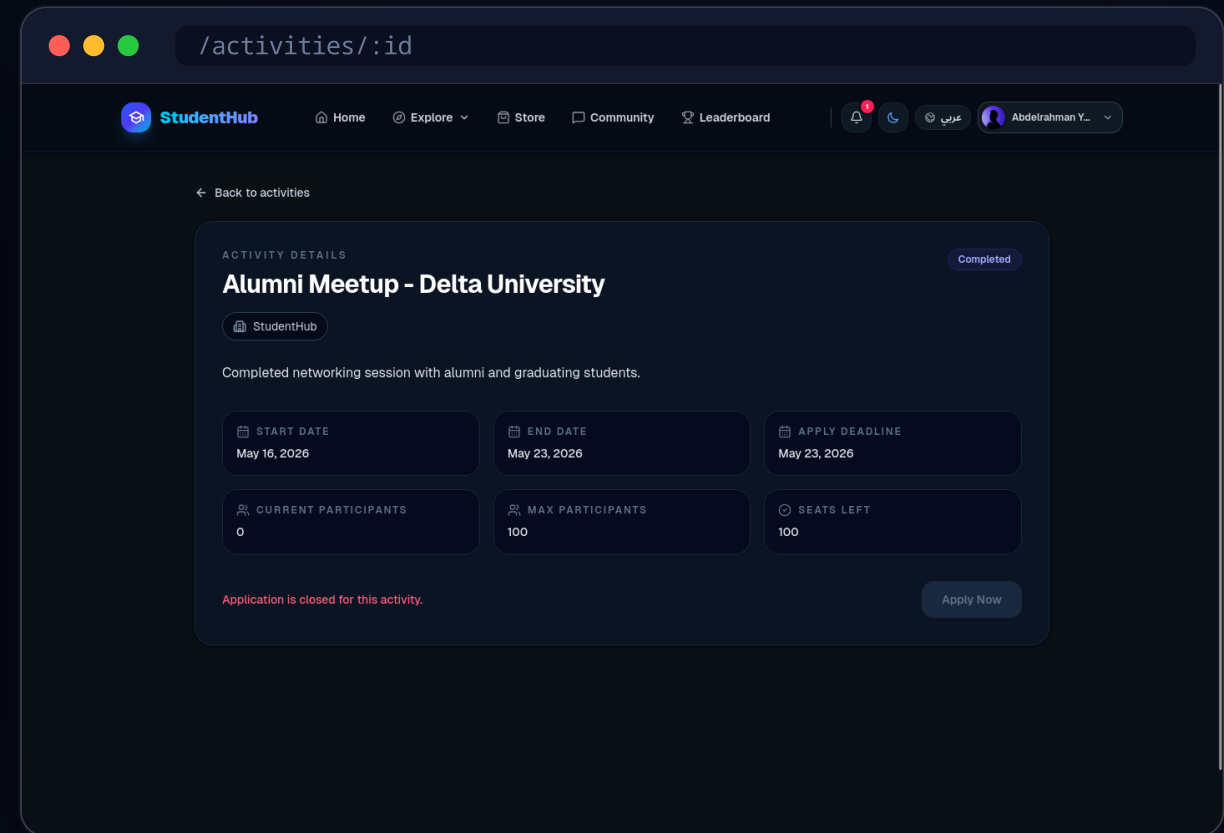
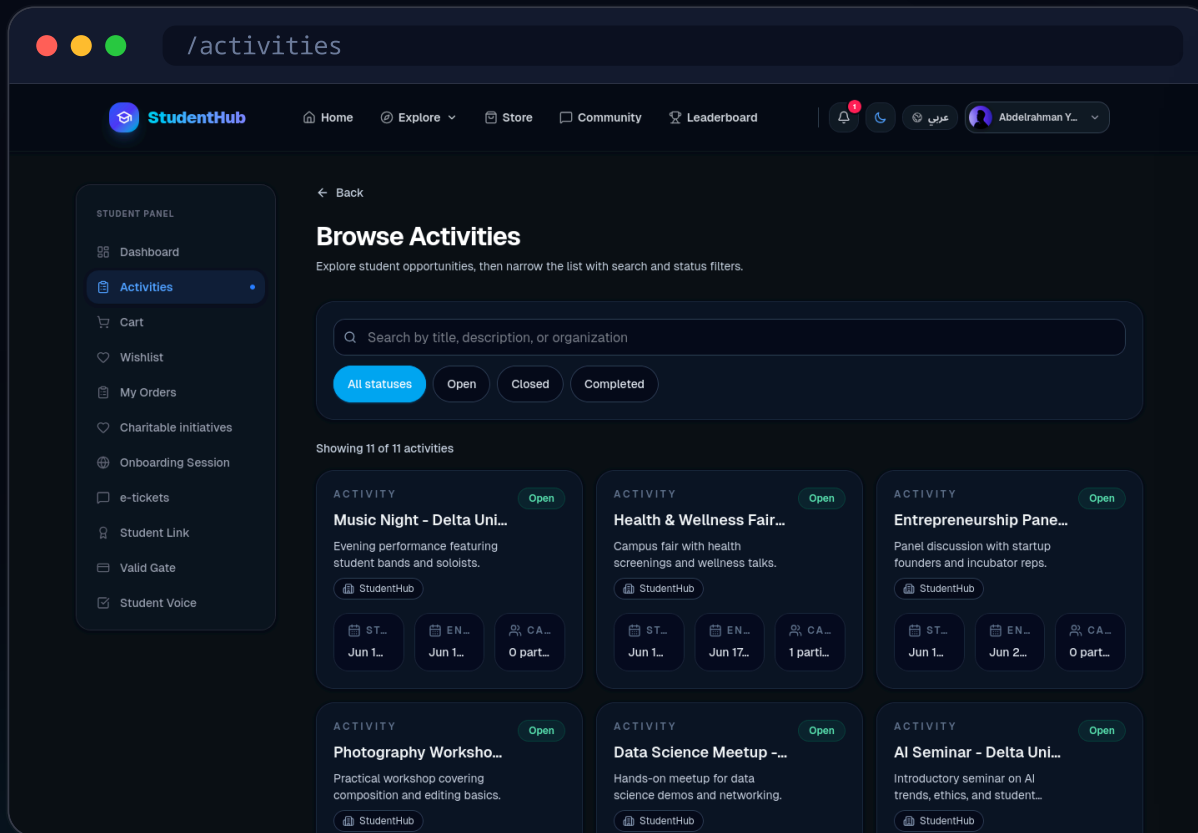
University & faculty communities with threads, polls, and discussions.



Store & Services

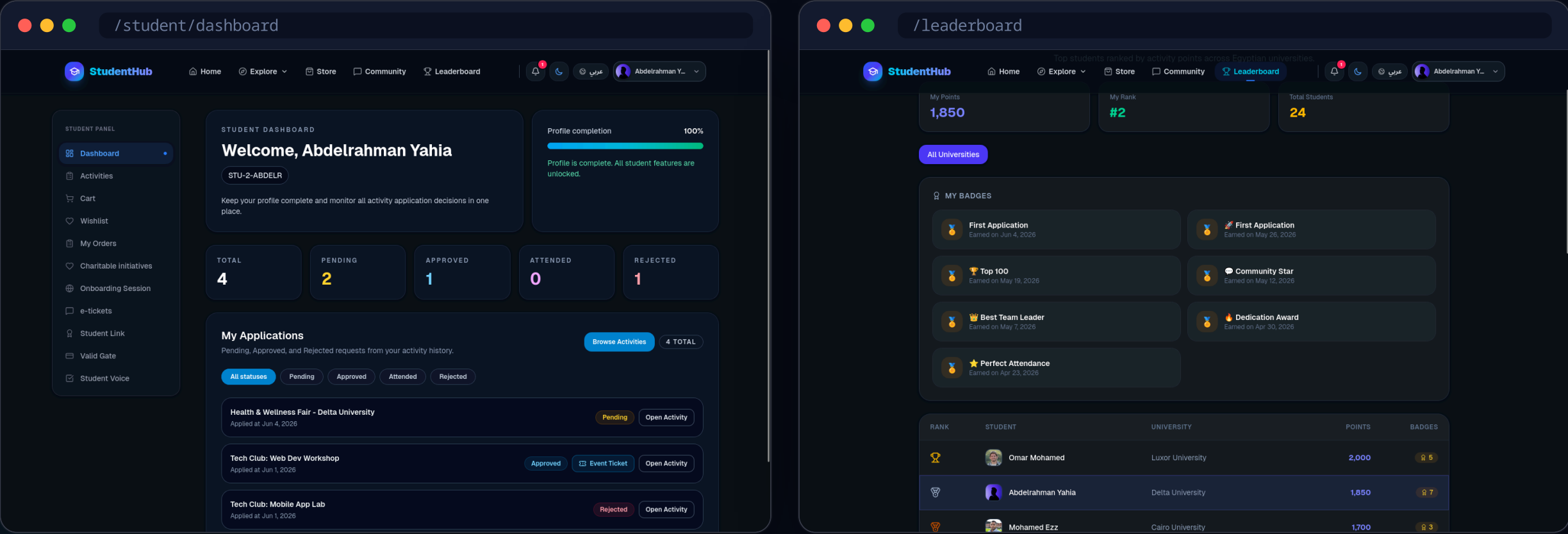
Campus store, food pre-orders, housing, scholarships, and charity.

Discover & apply in a few clicks



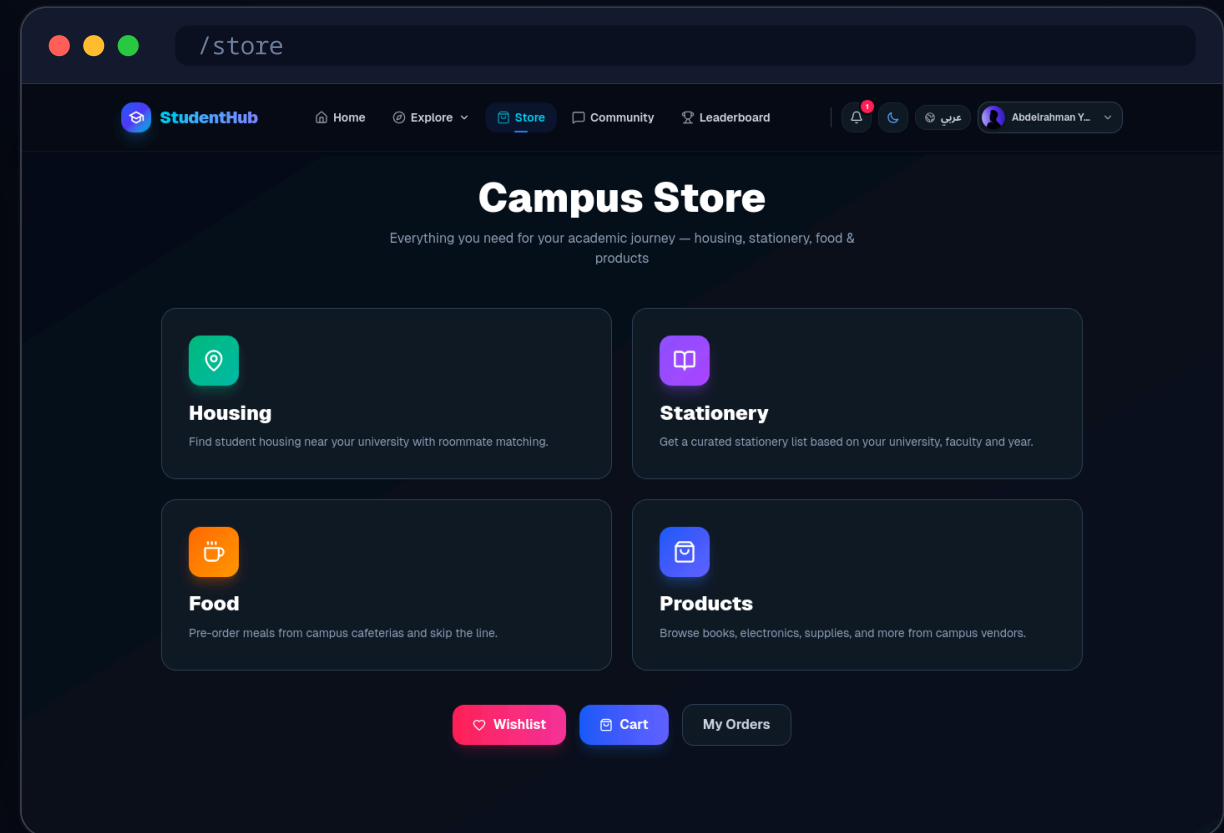
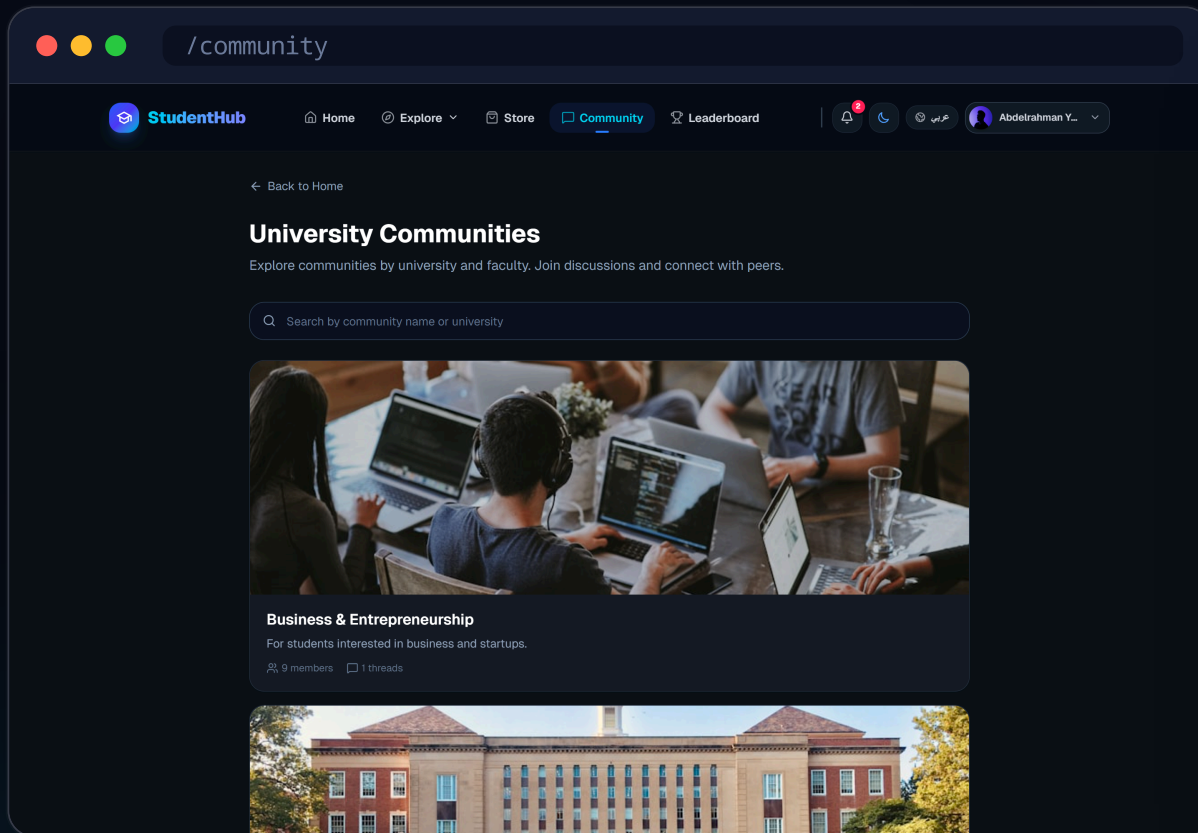
Search & filter activities → open details → **apply and reserve a seat**, with live capacity and deadlines.

Participation that counts



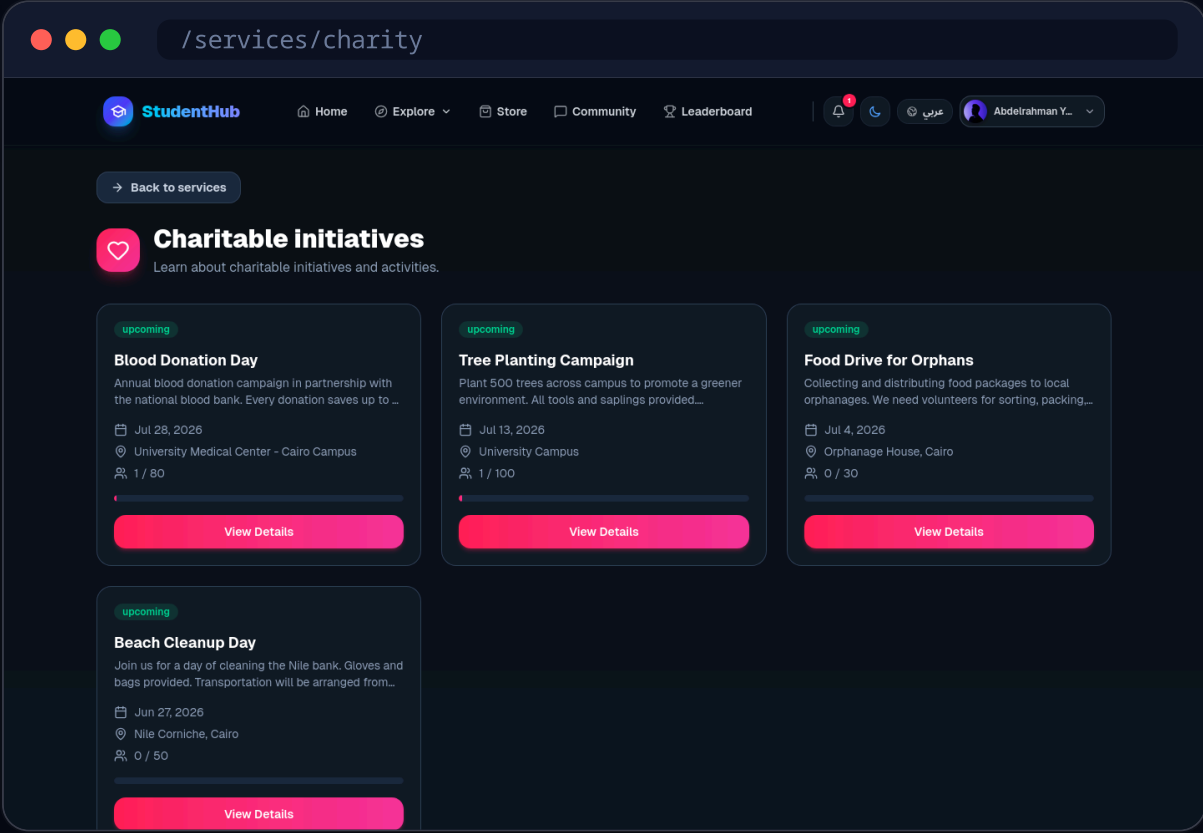
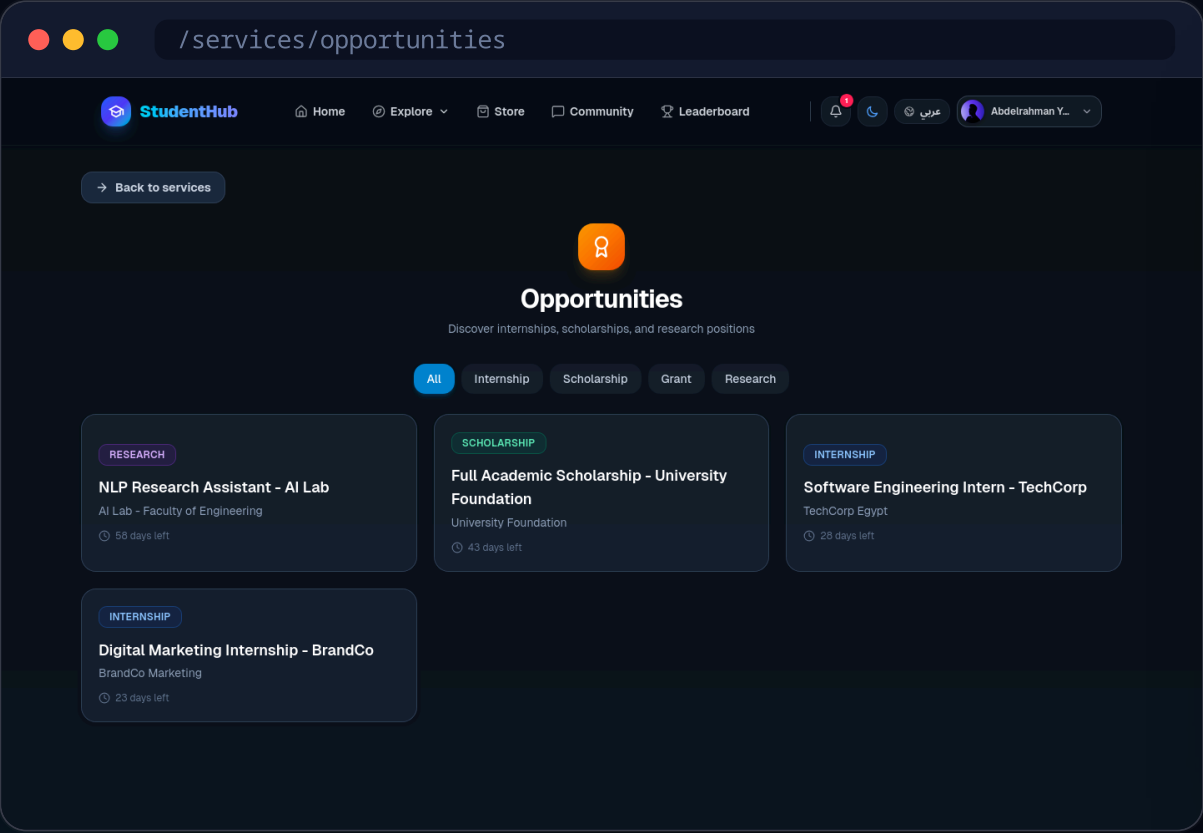
A personal dashboard tracks every application, while **points, badges, and leaderboards** reward real engagement.

A place to connect and shop



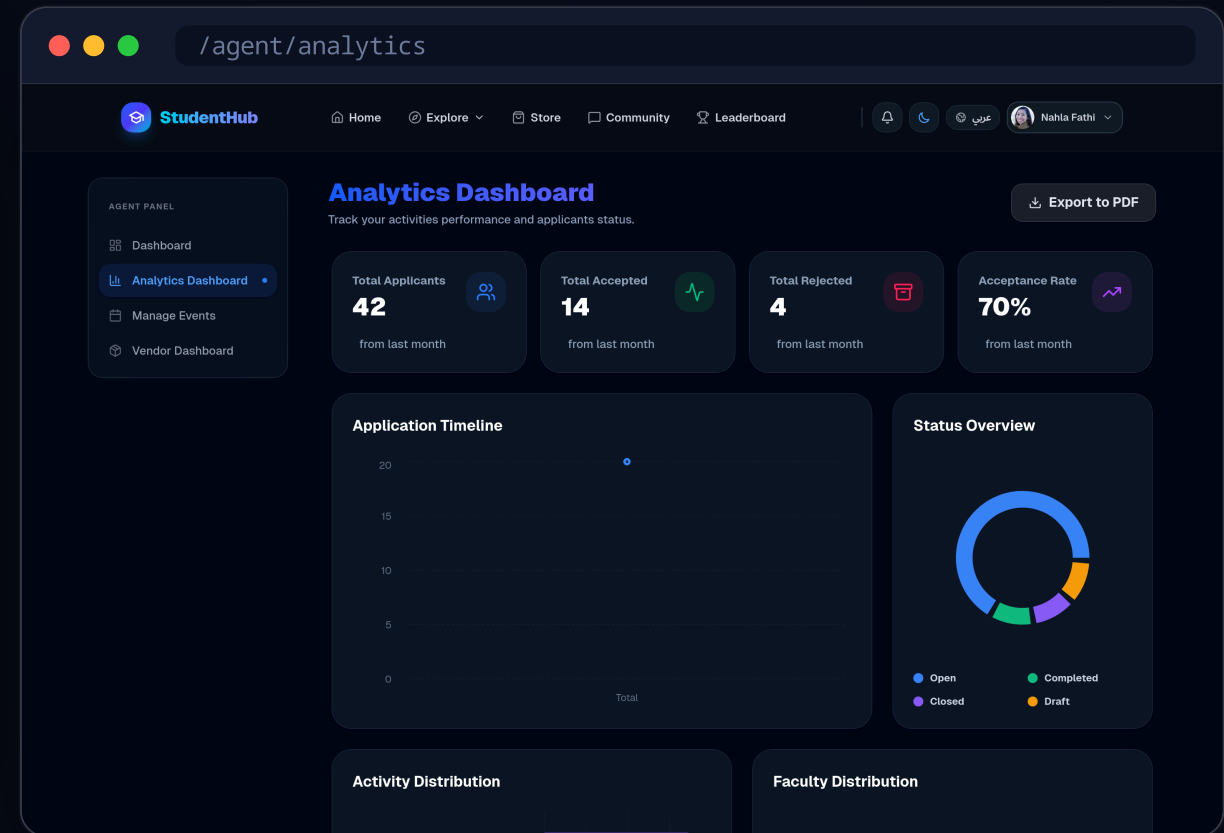
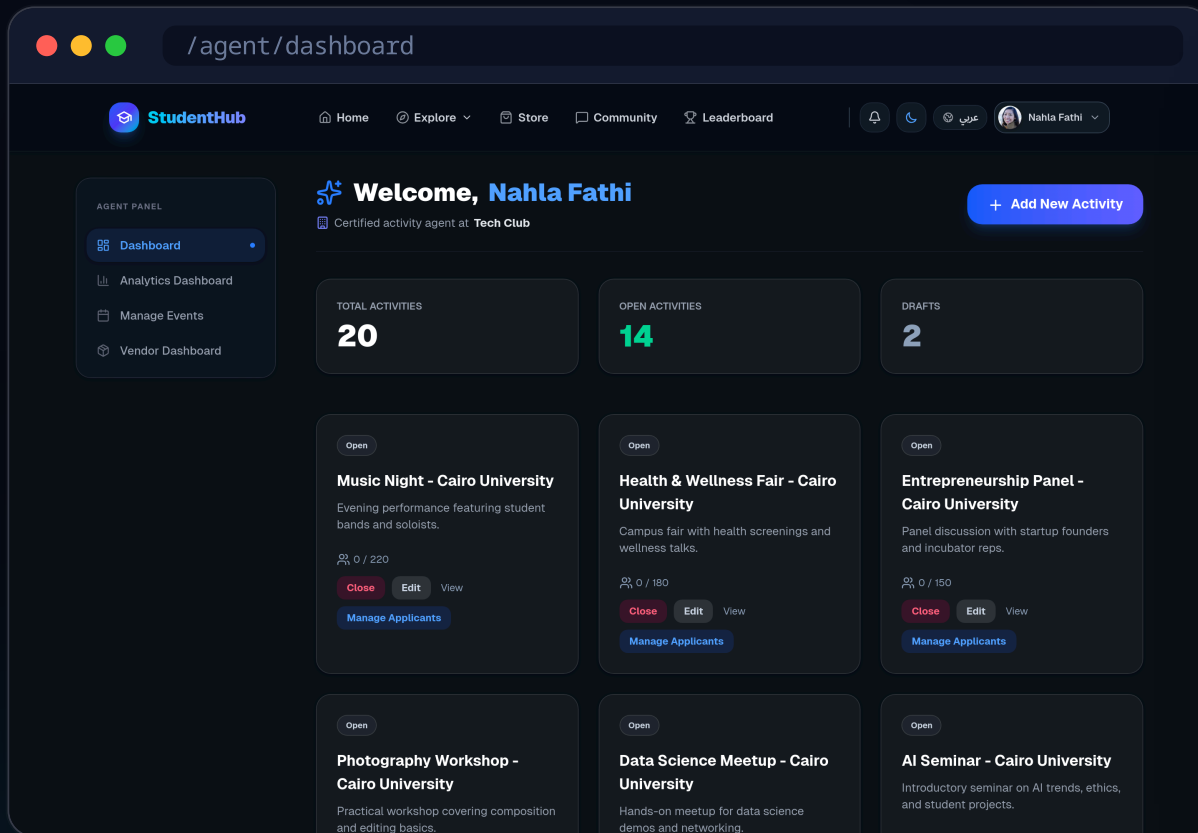
University & faculty **communities** for discussion, plus a **campus store** for housing, stationery, food, and products.

Beyond activities: real campus life



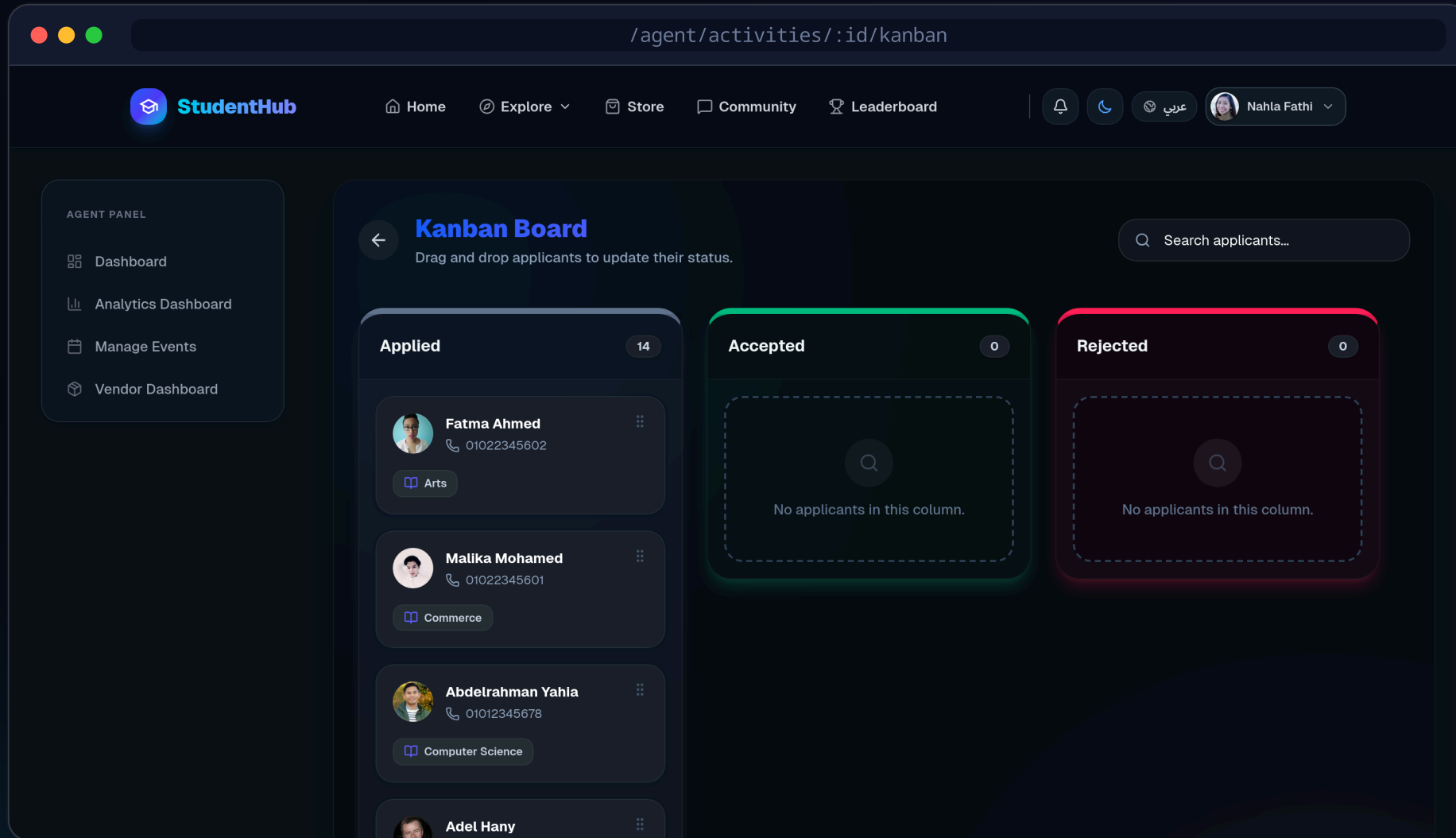
Internships & scholarships, charitable initiatives, polls, and onboarding sessions — all in one hub.

Run activities like a product team

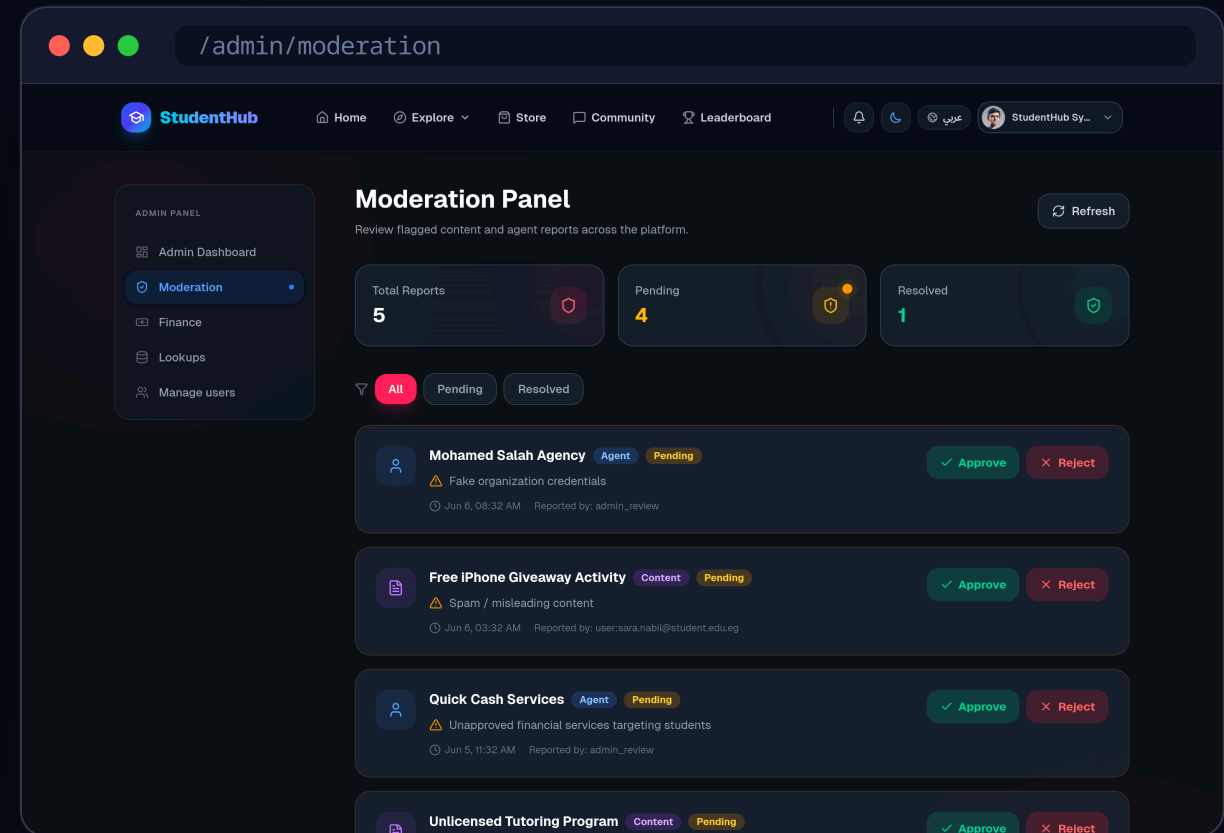
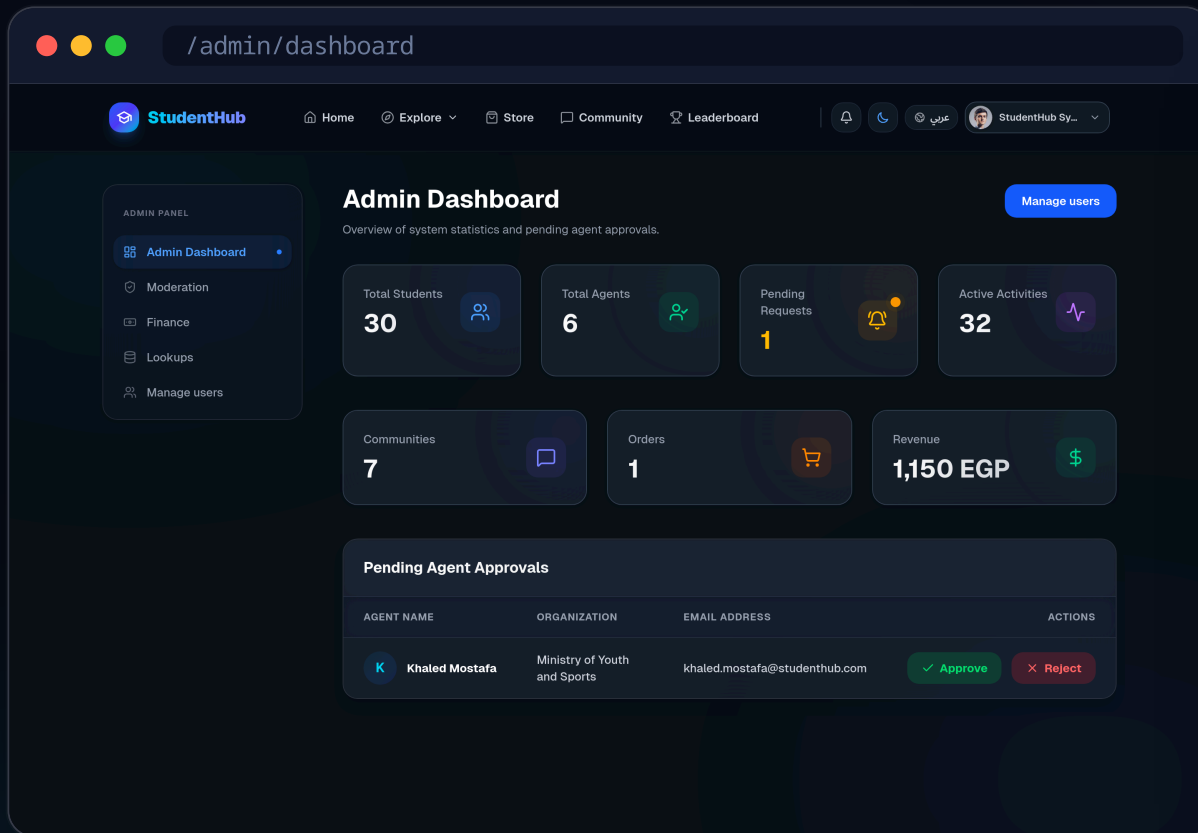


Manage every activity from one dashboard, with **real-time analytics** — applicants, acceptance rate, and distributions.

Drag-and-drop applicant management



A full control center



Approve agents, **moderate flagged content**, oversee **finance & revenue**, and manage users — platform-wide.

ACCESSIBILITY

Fully **bilingual** — English & Arabic with RTL



- Complete **Arabic / English** localization across every screen
- Proper **right-to-left (RTL)** layout mirroring
- Light & **dark mode** throughout
- Built for **Egyptian universities** from day one

UNDER THE HOOD

Clean, layered architecture

CLIENTS

React web app + **Flutter** mobile app — same API, one source of truth

▼ HTTPS · JWT Bearer tokens ▼

API LAYER

ASP.NET Core REST API · 17 controllers · role-based authorization

APPLICATION / DOMAIN Business logic, services & entities — framework-independent **Clean Architecture**

INFRASTRUCTURE **EF Core** + **SQL Server** · Cloudinary image storage · email & auth

Modern, production-grade stack



Web



React



Vite



Tailwind CSS



React Router



Mobile



Flutter



Dart



Cross-platform



Backend



.NET Core



EF Core



SQL Server



JWT



Cloudinary

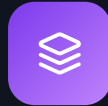
Three platforms, **one shared REST API** — documented with Swagger / OpenAPI.

The hard parts, **done right**



RBAC + JWT

Three roles with account-status gates (Pending / Active / Rejected) enforced on every endpoint.



Clean Architecture

Domain, Application, Infrastructure, API — testable and independent of frameworks.



Gamification engine

Points, badges, and leaderboards computed from real participation.



Media pipeline

Cloudinary-backed image uploads for profiles, activities, and the store.



QR check-in

Agents verify event attendance on the spot via QR scanning.



i18n + RTL

Full Arabic/English localization with mirrored right-to-left layouts.

HOW WE BUILT IT

A real GitHub workflow



Feature branch



Pull request



Code review



CI checks



Merge to main

500+

COMMITTS

135+

PULL REQUESTS

25

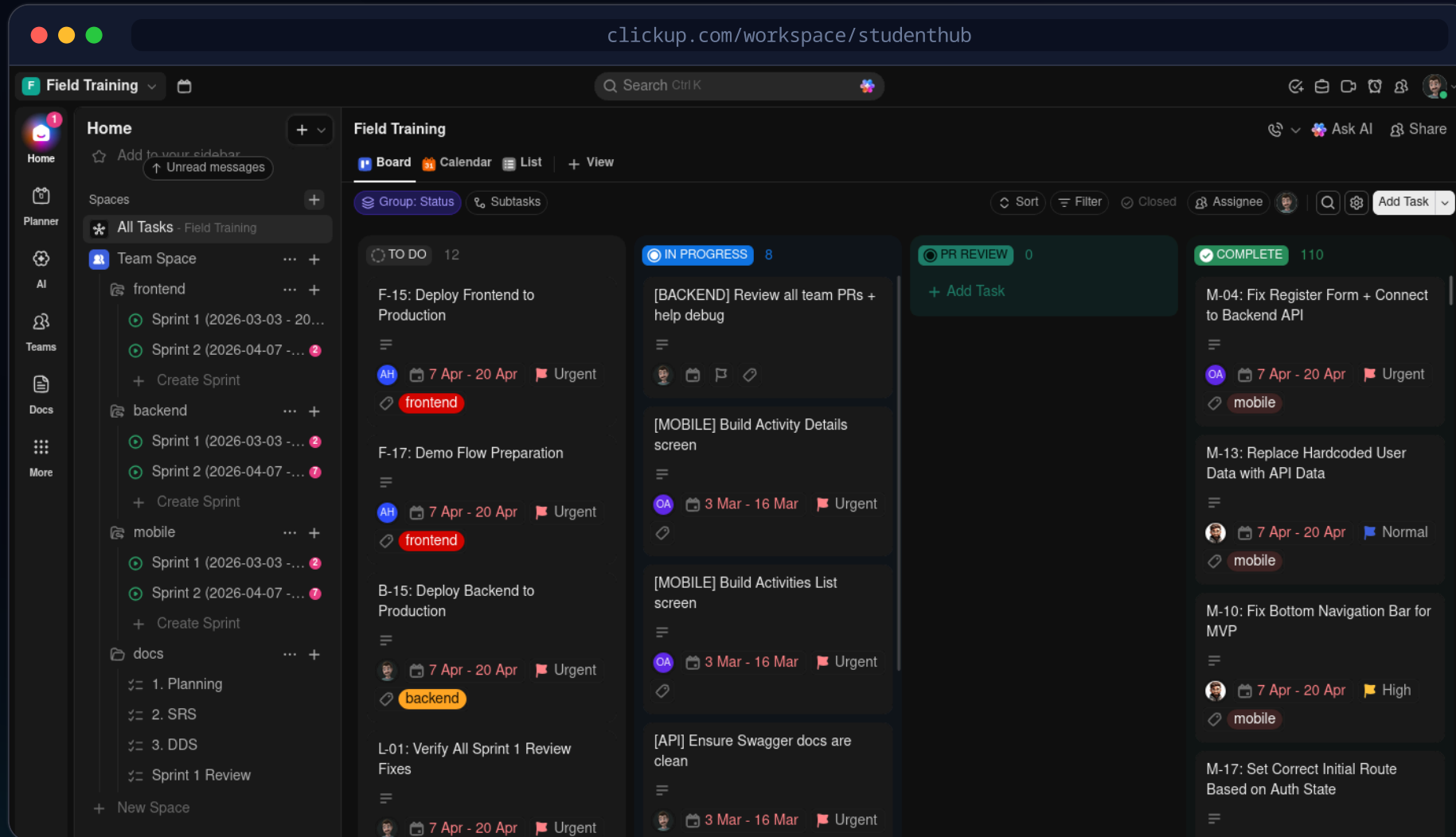
ISSUES TRACKED

~4 mo

FEB → JUN 2026

Conventional branch naming (feat/ · fix/ · chore/), reviews on every PR, and continuous integration.

Sprint execution via ClickUp



THE TEAM

Eight people, **clear ownership**



Abdelrahman Yahia

LEAD · BACKEND



Yousef Alaa

BACKEND



Mohamed Ezz

BACKEND



Marwan Mohamed

BACKEND



Omar Mohamed

FRONTEND



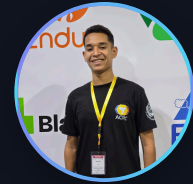
Ali Haggag

FRONTEND



Mahmoud Abdelhady

MOBILE



Omar Abdeen

MOBILE

4 backend · 2 frontend · 2 mobile — collaborating through pull requests & reviews.

BY THE NUMBERS

What “done” looks like

3

PLATFORMS · WEB · MOBILE ·
API

67

FUNCTIONAL REQUIREMENTS

110+

REST ENDPOINTS · 17
CONTROLLERS

47

WEB PAGES / SCREENS

3

USER ROLES

7

FEATURE MODULES

EN / AR

BILINGUAL + RTL

8

TEAM MEMBERS

What we **solved** along the way



Integrating many modules

Challenge: activities, store, community & services on one data model.

Solution: Clean Architecture with clear bounded contexts and a shared API.



8 people, one codebase

Challenge: avoiding conflicts across web, mobile & backend.

Solution: feature branches, pull-request reviews, and CI on every change.



Bilingual + RTL

Challenge: mirroring the whole UI for Arabic.

Solution: a central i18n layer driving direction and translations everywhere.

WHAT'S NEXT

Roadmap & future work

- **Online payments** & checkout for the store and tickets
- **Push notifications** for application decisions & events
- **Expanded mobile app** reaching feature parity with web
- **Advanced analytics** & exportable reports for admins
- **Calendar & reminders** synced to student schedules
- **AI recommendations** for relevant activities & opportunities



Thank you

We're happy to take your questions and give a live walkthrough.

github.com/3bdo-Yahya/Student-Hub